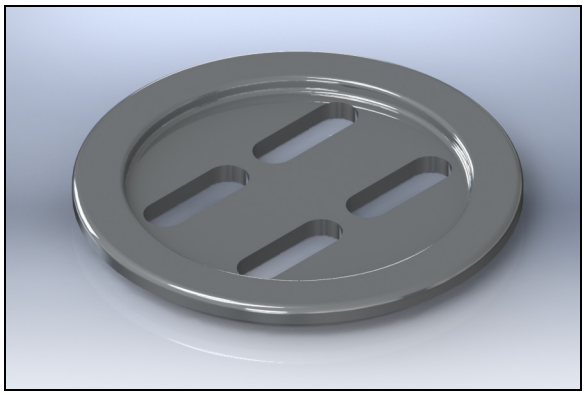
Active Learning Exercise, Part 1 — Creating a Basic Part

The first part created will be the `Binding Anchor` shown at right. We will use SolidWorks to create this part.

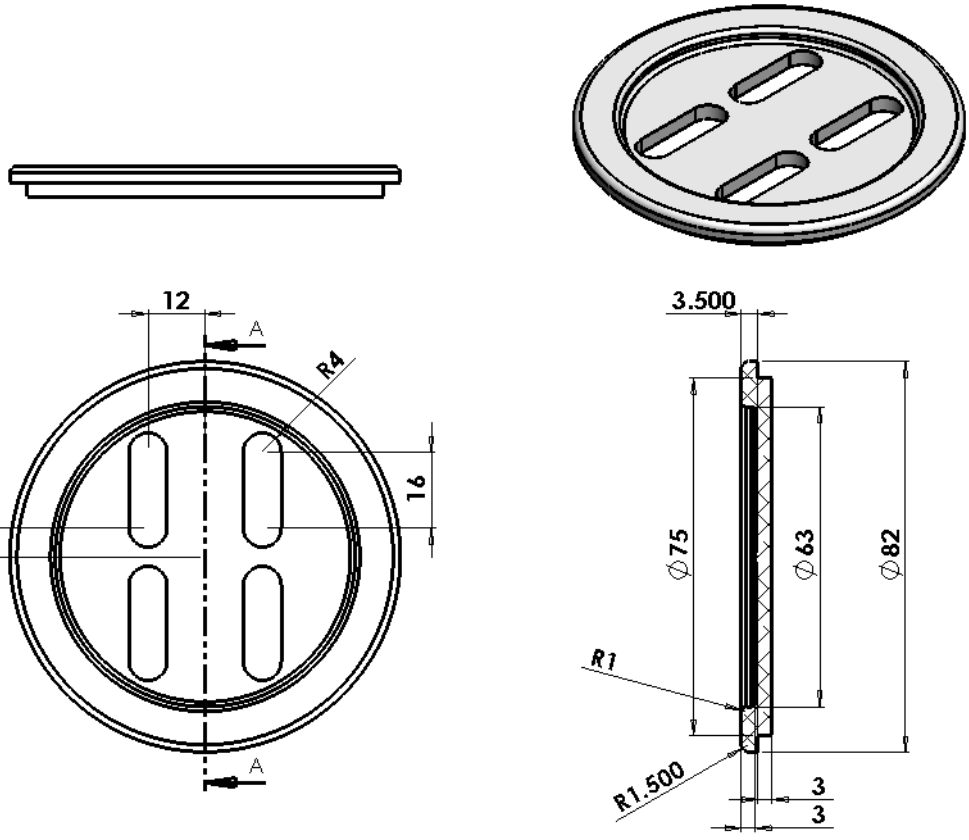


Design Intent

Before starting on the actual steps to create the `Binding Anchor`, we need to determine the design intent. This is a list of requirements the finished part needs to meet. The design intent will tell us what the finished part must be able to do.

- The `Binding Anchor` will position the binding on the deck.
- The `Binding Anchor` must allow the binding to be positioned both along the centerline of the deck as well as adjusting the angle to the deck to allow the rider to set a comfortable stance.
- The `Binding Anchor` clamps the binding to the deck.
- There must be no sharp edges to injure a rider.

The `Binding Anchor` will look like the drawing below. Step-by-step instructions are given below.



Task 1— Create a New Part Document

1 Create a new part. Click **New** on the Standard toolbar.

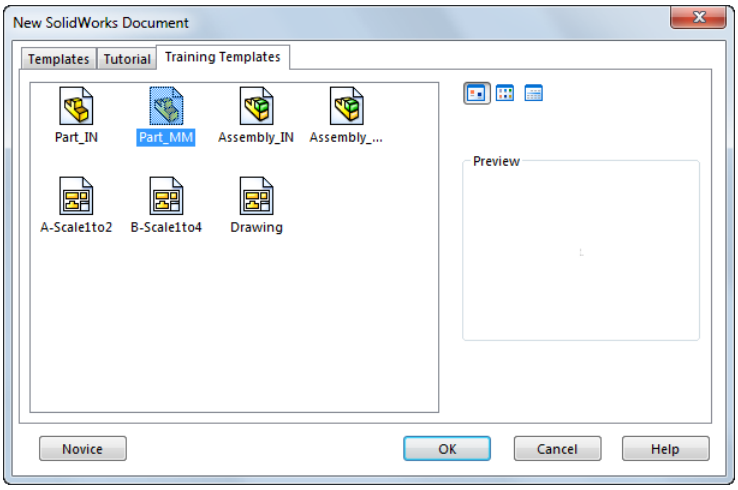
The **New SolidWorks Document** dialog box appears.

2 Click the **Training Templates** tab.

3 Select the **Part_MM** icon.

4 Click **OK**.

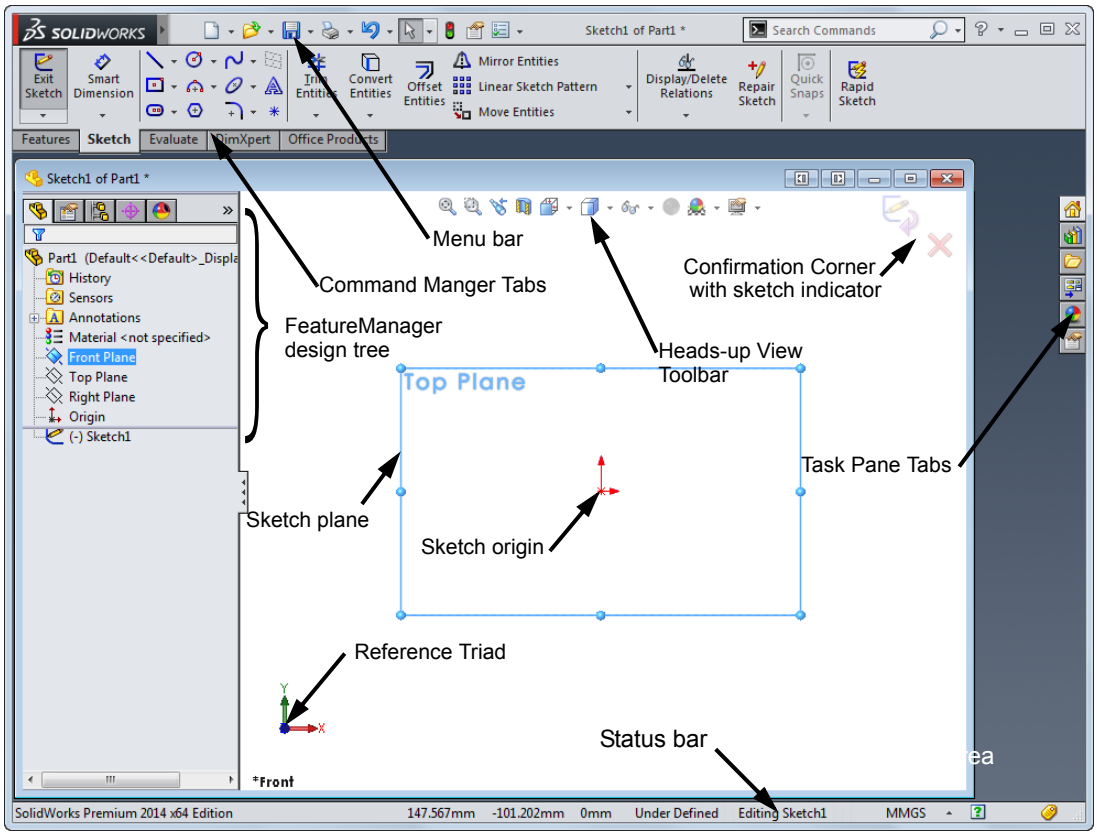
A new part document window appears.



Overview of the SolidWorks Window

When you create a new sketch:

- A sketch origin appears in the center of the graphics area.
- "Editing Sketch" appears in the status bar at the bottom of the screen.
- `Sketch1` appears in the FeatureManager design tree.
- The status bar shows the position of the pointer, or sketch tool, in relation to the sketch origin.



First Feature

The first feature requires:

- Sketch plane – Top
- Sketch profile – 2D Circle
- Feature type – Extruded boss feature

Sketching verses Drawing

The basis of most SolidWorks features is the sketch. Sketching is different from drawing in that drawings are created to the correct size as the lines and circles are drawn on the screen. With sketches, you only get the lines and circles close to their correct size. Dimensions and relationships will be added to make the sketch the correct size.

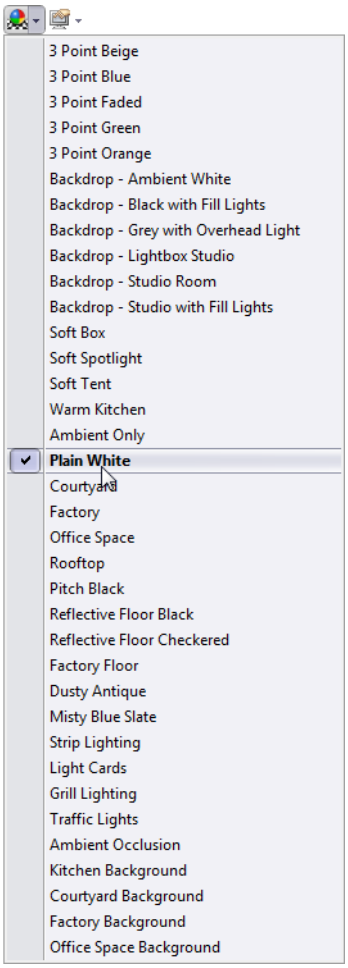
Open a Sketch

5. In the FeatureManager design tree, select (click once) the Top plane.
6. Select the **Sketch** tab on the Command Manager.
7. Open a 2D sketch. Click **Sketch**  on the Command Manager.

   The sketch opens on the Top plane.

Background

To make the images in this course easier to read, we will use a white background instead of the default. To change the background scene, click **Apply Scene** on the **Heads-up View** toolbar and select the scene you want to use.



View Orientation

When we open a sketch for the first feature, SolidWorks will automatically change the view orientation to be normal to the sketch plane. This makes it easier to see the sketch. It is like looking straight down on a piece of paper.

Confirmation Corner

When many SolidWorks commands are active, a symbol or a set of symbols appears in the upper right corner of the graphics area. This area is called the **Confirmation Corner**.

Sketch Indicator

When a sketch is active, or open, a symbol appears in the confirmation corner that looks like the **Sketch** tool. It provides a visual reminder that you are active in a sketch. Clicking the symbol exits the sketch saving your changes. Clicking the red X exits the sketch discarding your changes.

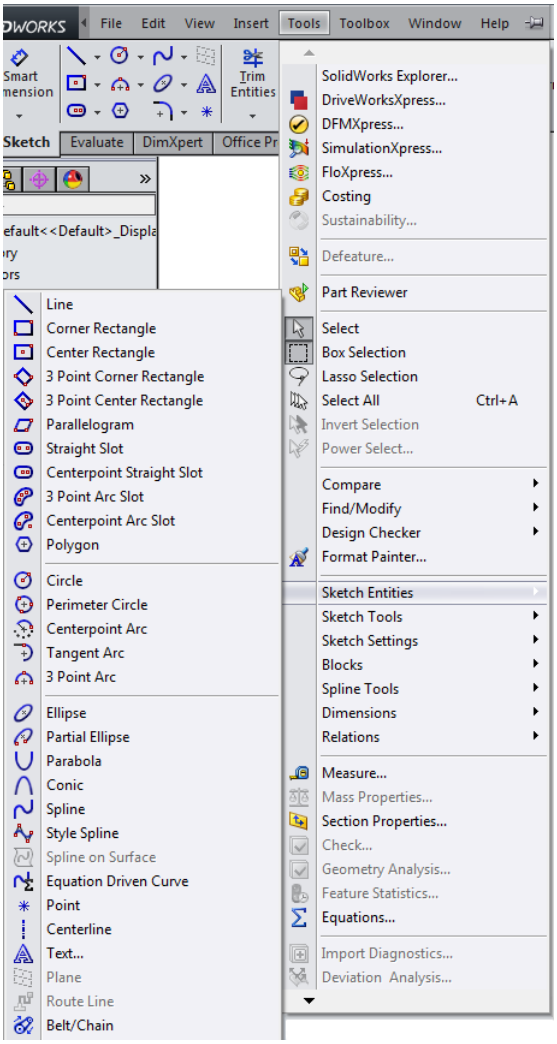
When other commands are active, the confirmation corner displays two symbols: a check mark and an X. The check mark executes the current command. The X cancels the command.

Sketch Entities

SolidWorks provides a variety of tools to create sketches. They can be found on both the **Sketch Entity** menu and most can also be found on the **Sketch** toolbar.

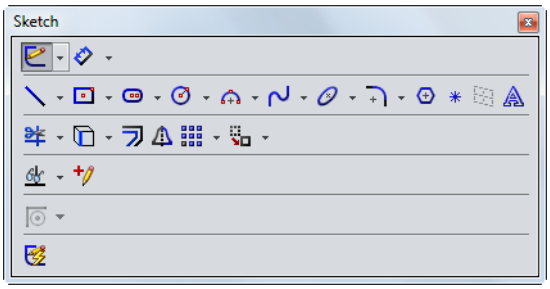
Sketch Menu

The Sketch Tools menu is found by clicking **Tools, Sketch Entities**. All of the sketch tools are listed in the menu.



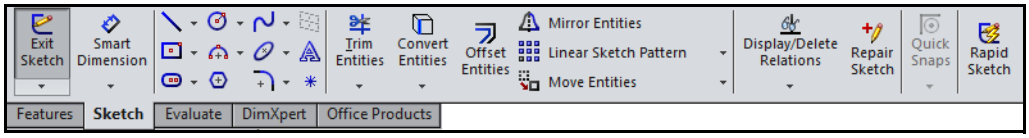
Sketch Toolbar

The Sketch Toolbar contains most of the sketch entities. It can be customized by adding or removing buttons.



Command Manager

Selecting the Sketch tab on the Command Manager will display the sketch tools.



Task 2 — Create the first sketch

The first feature will be a short cylinder, 75mm in diameter and 3.5mm thick.

The Circle

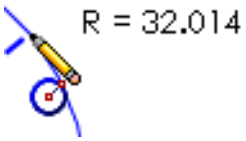
The circle tool creates 2D circles. Using the mouse, press the left mouse button at the location for the center of the circle, then (holding down the left mouse button) drag until the circle is approximately the correct size. Release the left mouse button.

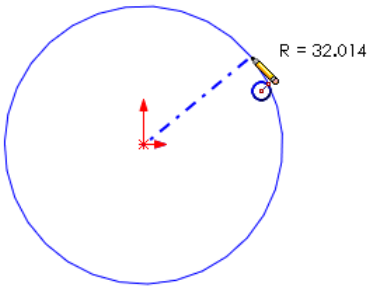
1 Click the **Circle** tool from the Sketch toolbar. The curser will show that the circle tool is active by displaying a circle under the drawing tool .

2 Move the cursor over the origin until an orange circle appears. The little yellow icon below the drawing tool will show that we are going to make the center of the circle coincident with the sketch origin.



3 Press the left mouse button and drag the circle until it is just about 32 mm. The cursor feedback will show the radius of the circle.



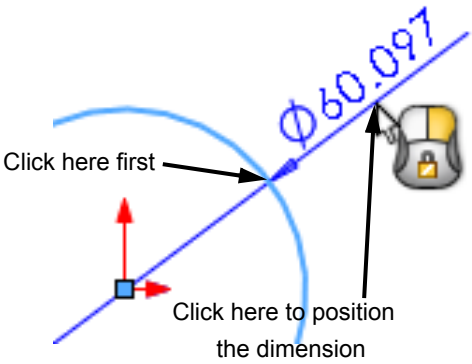


Dimension the sketch

To make the circle the correct size, we will add a diameter dimension to the sketch.

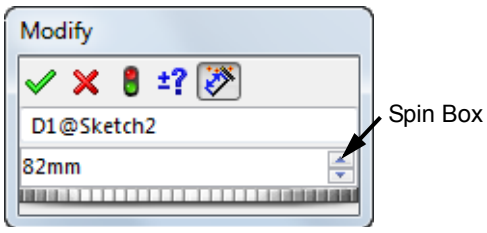
4 Click **Smart Dimension** on the **Sketch** toolbar. The cursor will look like this, indicating that the dimension tool is active.

5 Click on the circle, then move the cursor to the right and up on the screen. The preview dimension and witness lines will be visible. Click to set the dimension location.



6 Input the dimension by typing **82** in the Spin Box. Click to accept the dimension.

Spin boxes are used to input numerical data. They are called spin boxes because the numbers can be spin up or down using the arrows on the right.



Sketch Status

Sketches are normally fully defined before creating a feature with them. To be fully defined, the sketch geometry must be geometrically defined and positioned.

- To be geometrically defined, there must be enough dimensions and/or relationships to keep the size and shape of the sketch from changing if we try to drag it.
- To be positioned the sketch must also have dimensions or relationships that keep it from moving.

Sketch Color

The color of the sketch entities shows the status of the individual entity.

Blue - Under defined

Black - Fully defined

Red - Over defined

Extrude

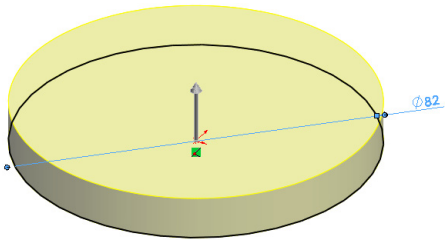
Once the sketch is completed, it can be extruded to create the first feature. There are many options for extruding a sketch including the end conditions, draft and depth of extrusion, which will be discussed in more detail in later lessons. Extrusions take place in a direction normal (perpendicular) to the sketch plane.

Task 3 — Extrude the first feature

Extruding the 2D sketch will produce a 3D solid. In this case, we will make a short cylinder.

1 Select the Features tab on the Command Manager.

Click **Extrude Boss/Base** on the Features toolbar. The model will reorient to the `Isometric` view and show a preview of the extrusion.

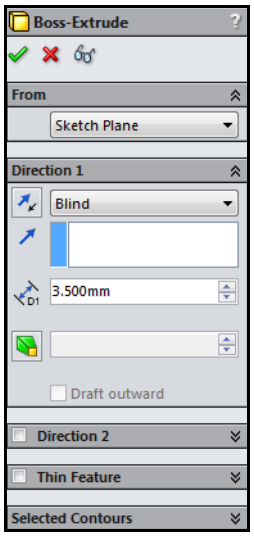


2 Preview graphics.

A preview of the feature is shown at the default depth.

A handle appears that can be used to drag the preview to the desired depth. The current depth of the preview can be seen in the PropertyManager.

3 In the PropertyManager, change the settings as shown.



- End Condition = **Blind**
- (Depth) = **3.5mm**

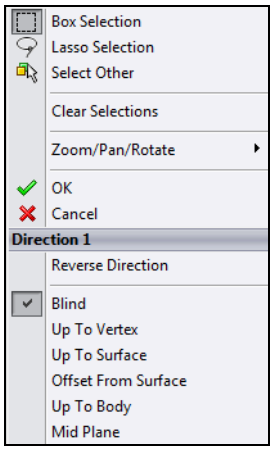
4 Create the extrusion. Click **OK**. . The extrusion now becomes a solid and a new feature, `Extrude1` is displayed in the FeatureManager design tree.

TIP:

The **OK** button on the PropertyManager is just one way to complete the command.

A second method is the set of **OK/Cancel** buttons in the confirmation corner of the graphics area.

A third method is the right-mouse shortcut menu that includes **OK**, among other options.

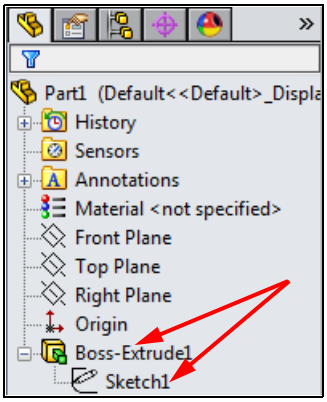


Blind Extrusions

Blind extrusions take the 2D sketch and move it, some specific distance, normal (perpendicular) to the sketch plane.

FeatureManager design tree

This extrusion is the first feature of our part. The FeatureManager design tree shows this feature by type and with a default name Boss-`Extrude1`.



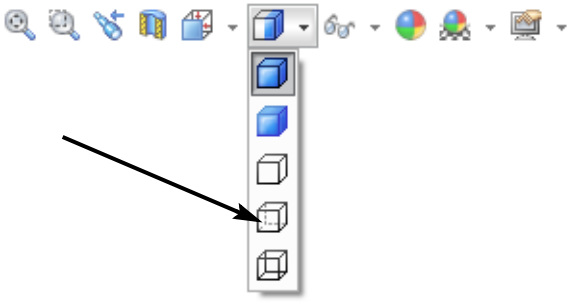
The sketch of the circle (Sketch1) is listed under the feature. It is said to be absorbed by the feature.

View Display

The View toolbar provides a quick method to change the way the model is displayed on the screen. It provides one set of tools to Zoom, Pan and Rotate the model view and another to change the way the model is displayed. In most cases, models are created in Shaded view because it most closely resembles the real world.

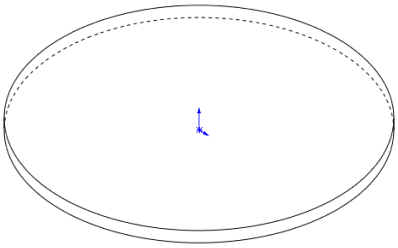


The Head-up View toolbar also provides the same functions. To change the view, click the **Display Style** pull-down list.



Change the display mode. Click **Hidden Lines Visible** on the **View** toolbar.

Hidden Lines Visible allows you to easily select hidden back edges of the part.



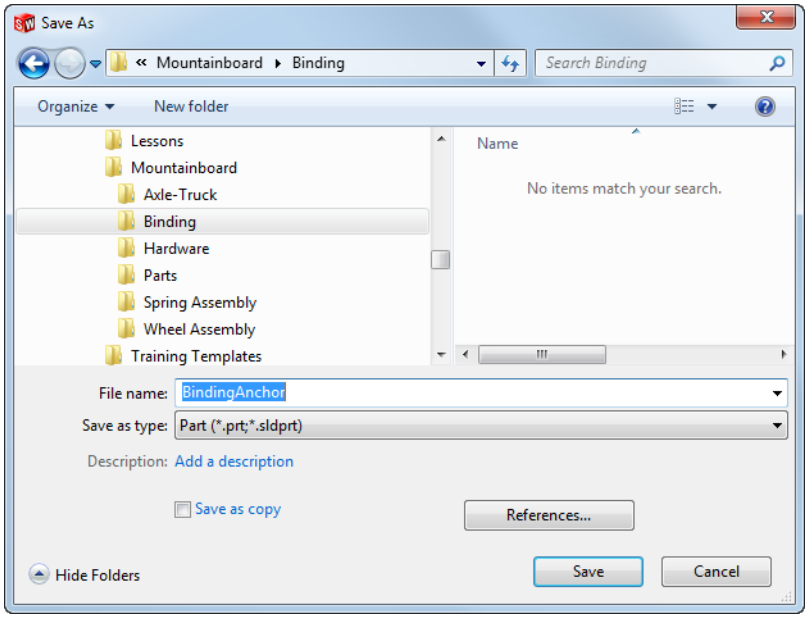
Save the Part

Save your work frequently. If you have a computer problem, you may loose everything you did since the last time you saved your work.

5 Click **Save** on the Standard toolbar, or click **File, Save.**

The **Save As** dialog box appears.

6 Type `Binding Anchor` for the filename.



7 Save the file to the folder `Binding` found under `SolidWorks Curriculum` and `Lessons_<year>\Mountainboard Design Project\Mountainboard`.

8 Click **Save**.

The `sldprt` extension is automatically added to the filename.

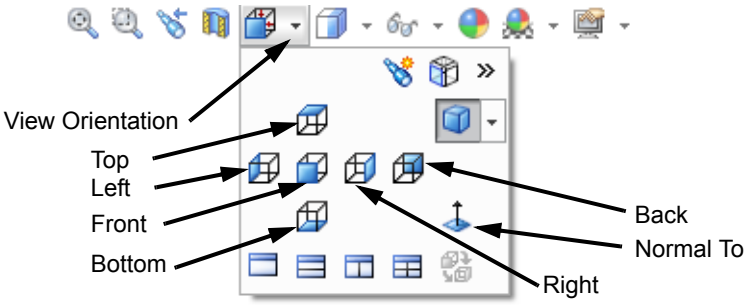
The file is saved to the current folder. You can use the Windows browse button to change to a different folder.

Note: All the files we create of the Mountainboard project should be saved in the appropriate folder under the folder `…\SolidWorks Curriculum` and `Lessons_<year>\Mountainboard Design Project`.

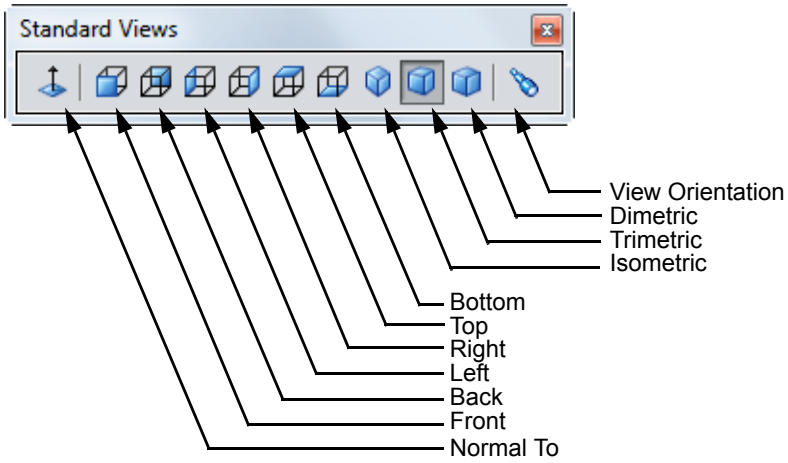
Changing views

The Standard Views toolbar or the Heads-Up View toolbar makes it easy to change your view of the model by simply clicking on the view you would like to see.

Heads-up View Toolbar

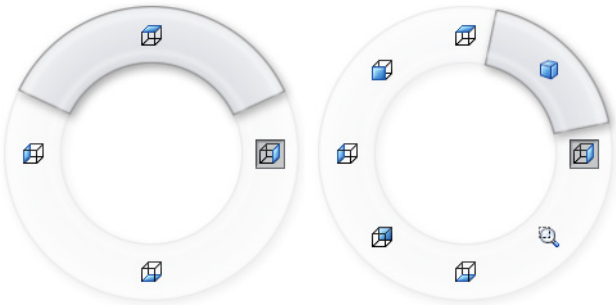


Standard Views Toolbar



Mouse Gestures

Mouse Gestures can also be used to access the different views. The tools available through Mouse Gestures can be customized to show either four or eight tools. The tools available will also depend on whether you are in a sketch, part, assembly or drawing. Shown are the default Mouse Gestures for a part.

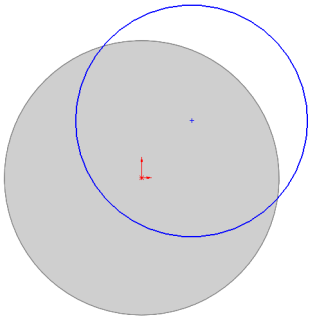


9 Change the view of the model to the **Bottom** view. Click on either the **Standard Views** or **Heads-up View** toolbar.

Task 4 — Add a second feature

The second feature will be another cylinder, slightly smaller than the first.

1. Change the display mode back to shaded. Click **Shaded With Edges** on the **View** toolbar.
2. Select the bottom face of the cylinder. It will turn blue to show that it is selected.
3. Start a new sketch by clicking the **Sketch** on the Sketch toolbar or pop up toolbar.
4. Select **Circle** on the Sketch toolbar.
5. Draw a circle, slightly smaller than the size of the cylinder. It does not have to be centered on the cylinder. The circle is blue, indicating that the sketch is Under Defined.



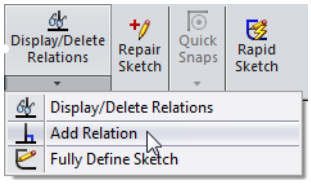
Task 5 — Adding sketch relationships

Sketch relationships are used to force a behavior on a sketch element to capture design intent. Some are automatic, others can be added as needed.

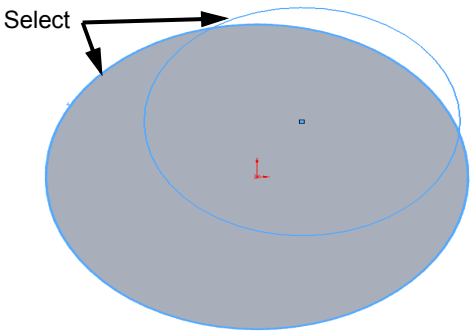
When adding relationships, only those relationships that are appropriate for the sketch entities selected will be shown in the PropertyManager.

1. Click **Add Relation** on the Sketch toolbar. **Add Relation** will appear in the PropertyManager.

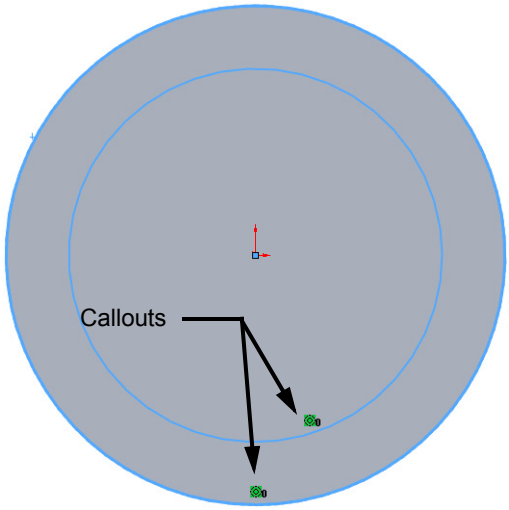
Note: When using the Command Manager, click **Display/Delete Relations**, then select **Add Relations** from the dropdown list.

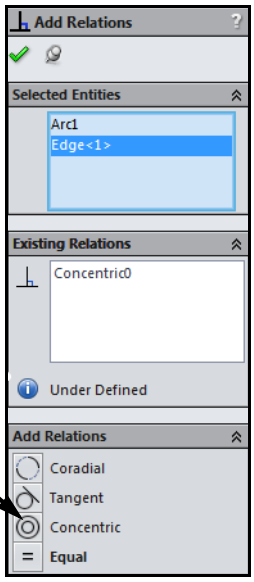


2. Select the circle and the edge of the cylinder.



3 Click **Concentric** in the **Add Relations** box. Then click **OK** . The circle will move to a position where it is centered on the cylinder. Callouts will show on the sketch to show the relationships.





Callouts

Callouts provide a display of existing conditions. The relationship callouts show which relationships exist and between which sketch entities.

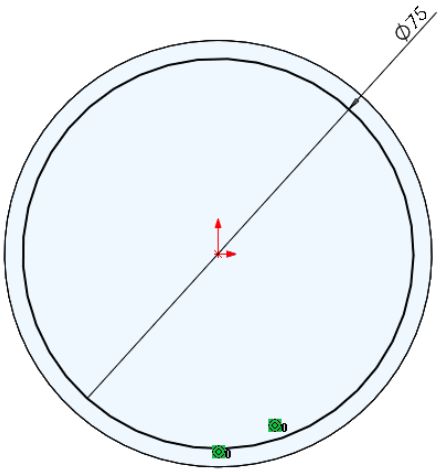
Task 6 — Dimension the circle

The concentric relationship defines the position of the circle, but it is still blue (under defined) because it doesn't have a dimension for its diameter.

1 Click **Smart Dimension** on the Sketch toolbar.

2 Click on the circle, move the cursor to the right the click again to set the dimension position.

3 Type **75** for the value. Click . The circle will now be black to show that it is fully defined.



Status Bar

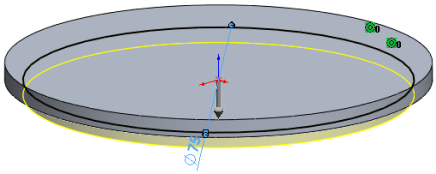
The status bar, located at the bottom right of the graphics window will also show the state of the sketch. Fully Defined

Change the viewpoint

When creating the first feature, our viewpoint was automatically changed to the `Isometric` view. After the first feature, we must change the view to best see the preview of the new feature.

Task 7 — Extrude the second feature

1. Change the viewpoint to `Dimetric` by clicking on the Views Orientation toolbar.
2. Click **Extrude** on the Features toolbar.
3. Select **Blind** for the type of extrusion.
4. Type **3mm** for the depth. Check the preview shown in yellow. It shows that material will be added to the bottom of the first cylinder.
5. Click .



Task 8 — Cut a recess in the top of the part

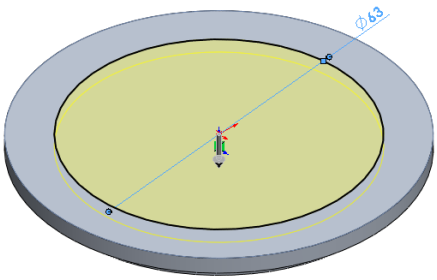
Material needs to be removed from the top of this part to:

- Reduce weight. Each part must be designed to be as light as possible so that the assembled mountainboard is not too heavy to be carried.
- Lower the tops of the screws used to bolt this part to the deck. This will reduce the chance of anything (pants leg, shoe laces, etc.) getting caught on the screw heads.

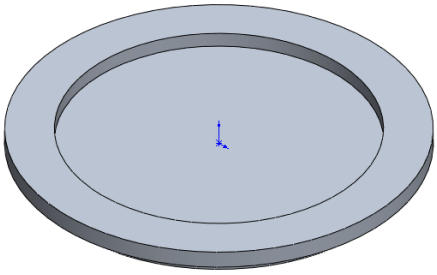
Removing material by extruding a cut

Material can be added or removed by extrusion. The process to add or remove material works the same in that you start by creating a 2D sketch. That sketch is then moved normal to the sketch plane. If you are creating a boss, the enclosed volume is added to the part. If you are creating a cut, the enclosed volume is removed from the part.

1. Orient the part to the `Top` view by clicking on the **Views Orientation** toolbar.
2. Select the top face of the model, and click **Sketch** to start a new sketch.
3. Click **Circle** on the Sketch toolbar.
4. Draw a circle from the center of the top face. Make its radius about **30mm**.
5. Dimension the circle to be **63mm** in diameter.
6. Reorient the model to the `Isometric` view.
7. Click **Extruded Cut** to use the circle to cut away some material.
8. Check the preview, by default, cuts go into the existing part.
9. Type **3mm** for the depth.



10 Click ✔. The **Cut Extrude** command removed a cylinder shaped volume from the part.



Calculating the weight of the part

In any design, it is important to keep track of the weight of each individual part. In the case of the Mountainboard, if the individual parts become too heavy, the total weight of the board may exceed a reasonable weight to be carried up the hill.

The weight of the part can be calculated by multiplying the volume of the part by the density of the material.

- Weight = Volume x Density

Calculate the volume

The total volume is the sum of the volumes created by the two extrudes minus the volume of the cut.

- Total Volume = Volume of each of the two extruded cylinders - volume of the extruded cut
- Volume of a cylinder = Area of the circle times the cylinder height = Pi times the diameter squared divided by 4, times the cylinder height = $(Pi * D^2/4)\ h$
- Total Volume = $(3.14 * 82^2 / 4)(3.5) + (3.14 * 75^2 / 4)(3) - (3.14 * 63^2 / 4)(3)$ = 18,483.56 + 13,253.60 - 9,351.74 = **22,385.42 cubic millimeters**

Find the density

The density of engineering materials can be found in many ways. There are numerous engineering handbooks or several sites on the internet. One such site is MatWeb (www.matweb.com).

The `Binding Anchor` will be made from 2015 Aluminum. Using MatWeb, the density for 2015 Aluminum is 2.8 g/cc. There are 1000 cubic mm in 1 cc, so the density would be:

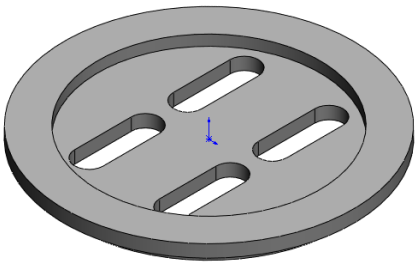
2.8 g/cc x .001 cc/mm = **.0028 g/mm^3**

Calculate the weight

Weight = Volume x Density = 22,385.42mm^3 x .0028 g/mm^3 = **62.68 grams (2.21 oz)**.

Task 9 — Create the screw slots

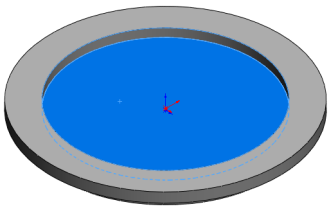
To make the position of the binding adjustable, the binding anchor will have four slots. These will allow the position of the bindings to be moved along the centerline of the deck.



The slots are symmetrical, so we will use a function called mirroring to make sure the slots always remain symmetrical if we later need to change their size.

Create a sketch

1. Select the face of the part created by the cut.



2. **Open** a sketch by clicking on the **Sketch** toolbar.
3. Change to the `Top` view by clicking on the **View Orientation** toolbar.

The Sketch Mirror tool

Mirror Entities and **Dynamic Mirror Entities** create symmetric relationships between sketch entities about a centerline. The **Dynamic Mirror Entities** command can be used *while sketching* and the **Mirror Entities** *after sketching* with the same final results.

Lines and Centerlines

The Line tool draws straight lines. If the line is vertical, the cursor will show a yellow callout to indicate that a **Vertical** relationship will be added. If the line is horizontal, the cursor will show an to indicate that a **Horizontal** relationship will be added.

Centerlines are construction geometry. They are used to position other entities but do not result in features.

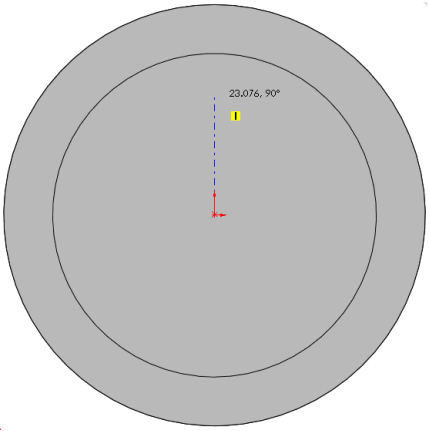
While you are sketching, the callouts will be yellow , indicating which relationship will be added when you release the mouse button. The green callouts show relationships that have been added .

Note: The color of the callouts indicating existing relationships, green in this case, can also be cyan. Their color depends on the color scheme chosen in the SolidWorks Options.

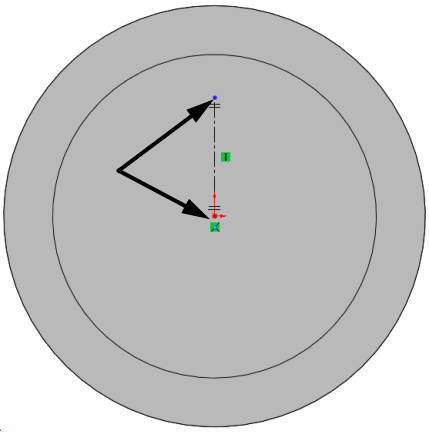
Create a centerline

4. Click **Centerline** on the Sketch toolbar.

5 Sketch a vertical **Centerline** from the sketch origin. The length is not important. Make sure the cursor displays a , indicating a **Vertical** relationship will be added.

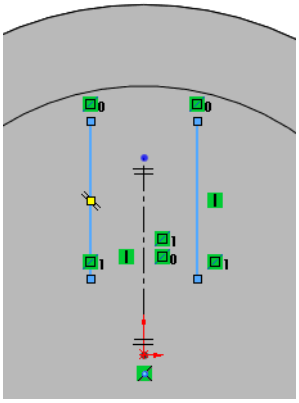


6 Click **Dynamic Mirror Entities** on the **Sketch** toolbar. A pair of parallel marks will appear at each end of the centerline to show that we are in the mirror mode.



7 Sketch a vertical line to one side of the centerline. As soon as you finish drawing the line, a mirror image will be drawn automatically on the other side of the centerline.

The callouts will show that a symmetric relationship has been added between the endpoints of each line.



Arcs

There are three tools provided to create arcs:

Tangent Arc — Adds a tangent relationship to the entity it is sketched from.

Center Point Arc — Defined by a center point and a radius.

3 Point Arc — Defined by two endpoints and a radius.

The choice of arc tools depends on the geometry that needs to be created.

Add an Arc

The sketch of the slot is composed of two straight lines and two arcs. The arcs must be tangent to both lines.

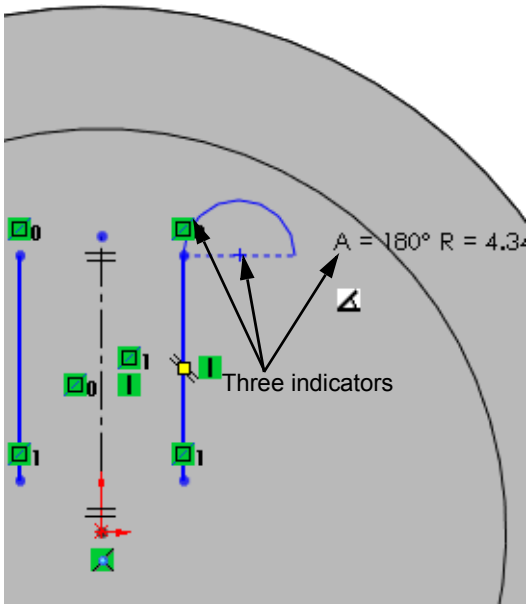
8 Click **Tangent Arc** the **Sketch** toolbar.

9 Place the cursor over the end of the right vertical line and drag an arc up and around to the right until you get cursor feedback showing you have gone 180 degrees.

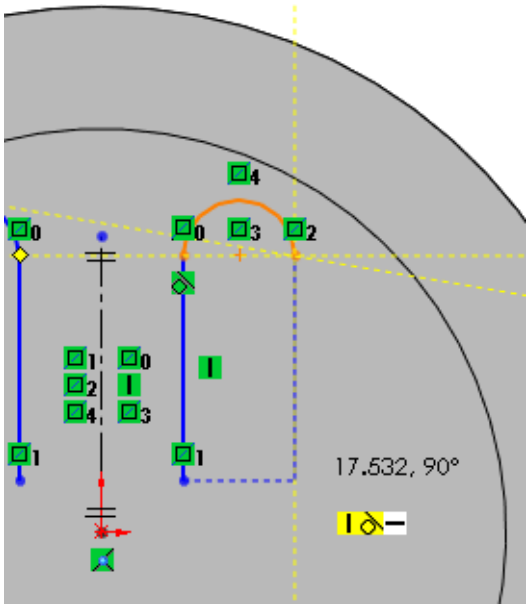
There will be three indicators that you have gone 180 degrees:

- A blue dashed line (inference line) from the center of the arc
- The angle symbol under the drawing tool
- The arc degree feedback (A=180)

When you release the mouse button, a second symmetric arc will be drawn automatically.

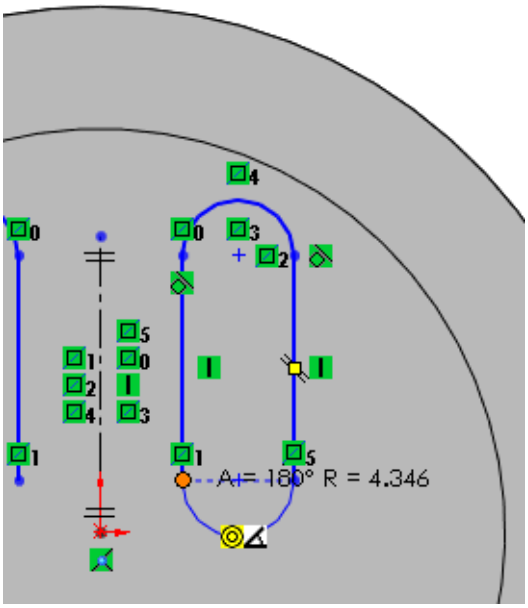


10 Draw another vertical line, from the end of the arc, vertically downward until you get an blue inference line from the bottom end of the first line.

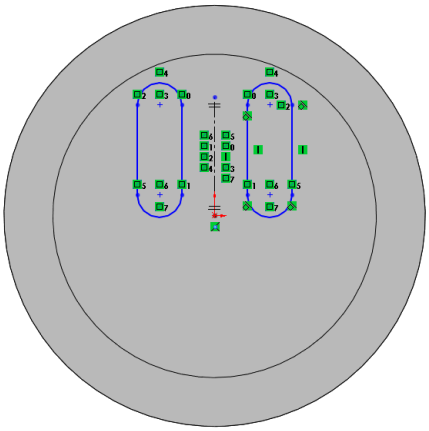


11 Finish the sketch with another **Tangent Arc**.

12 Turn off the sketch **Dynamic Mirror Entities** tool by clicking on the Sketch toolbar.



Review the progress



With sketch mirroring turned on, each entity we drew had a mirrored entity drawn on the other side of the centerline. The symmetric relationships added by the mirror tool will make these sketch elements retain the symmetry we desire.

Callouts show that the arcs are tangent to the two lines it connects to and symmetric to the other arcs and centerlines . The numbers next to each symmetric relationship show the pairs of symmetric elements.

With all the symmetric relationships, the number of callouts displayed may make the sketch elements hard to see, so we can turn them off.

Turn off the callouts

To toggle off the callout display of existing relationships, click **View, Sketch Relations** from the menu. This command is a toggle that turns the callouts on or off.

13 Click **View, Sketch Relations**.

14 Turn off the callouts by clicking **View, Sketch Relations**.

Mirror after sketching

Mirrored entities can also be created after creating sketch entities. To mirror after sketching, select the centerline about which you want to mirror and all the entities you want to mirror.

To mirror after sketching, click **Mirror Entities** .

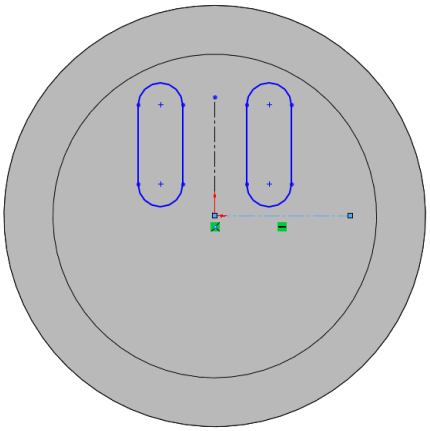
Task 10 — Mirror the two slot sketches

To complete the pattern of slots, mirror the two slots across a horizontal centerline.

1 Sketch a horizontal centerline from the origin to the right.

2 Click **Mirror Entities** on the Sketch toolbar.

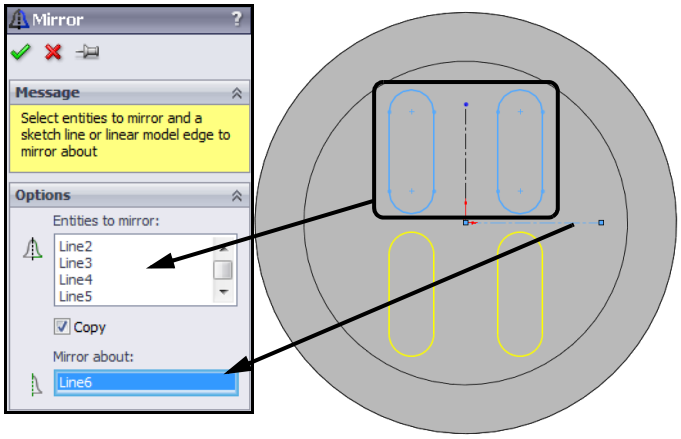
3 Turn off the **Centerline** tool by clicking on the tool again.



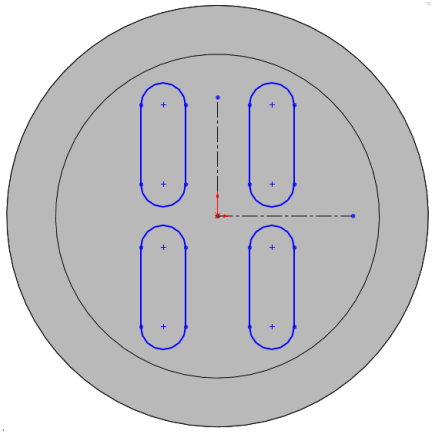
4 Select the four lines and four arcs as the **Entities to mirror**. Select the horizontal centerline for the entity to **Mirror about**.

Make sure you *do not* have the vertical centerline selected.

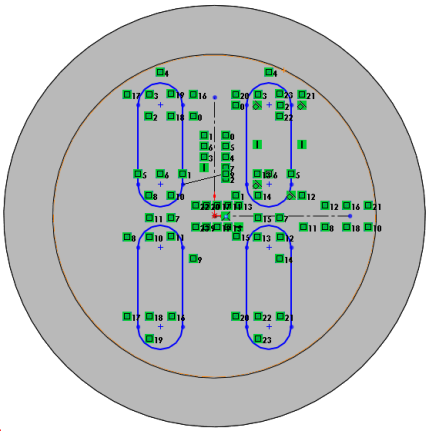
Click .



5 We now have a sketch that will cut the four slots.



View Relationships



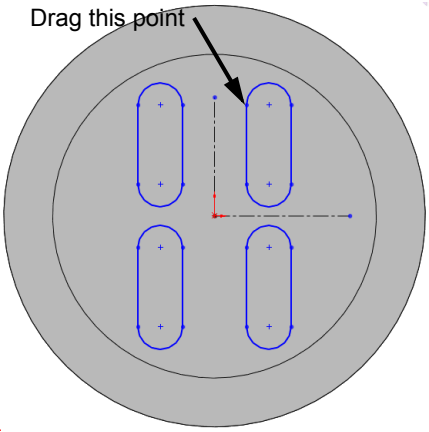
6 Click **View, Sketch Relations**.

The callouts show that the all the arcs are tangent to the two lines they connect to and symmetric to the other arcs and centerlines . The numbers next to each symmetric relationship show the pairs of symmetric elements.

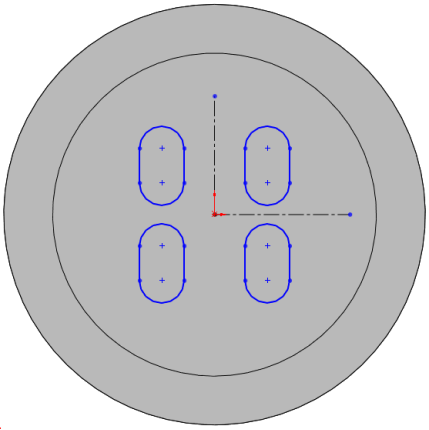
7 With this many symmetric pairs, the number of callouts can make it difficult to see the sketch. Turn off the callouts by clicking **View, Sketch Relations**.

Test the relationships

All four slots sketches should have symmetric relationships. Anything done to one slot should be mirrored into the other sketches.



8 Drag the point shown. All four slots should change shape together.

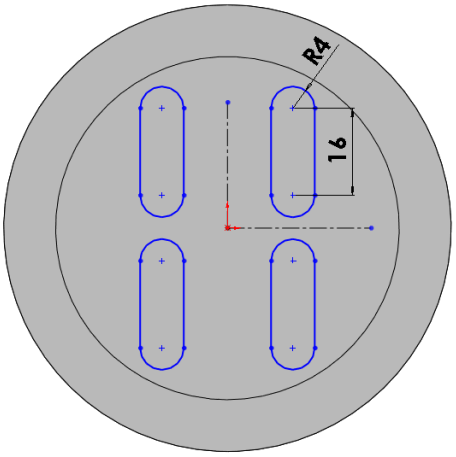


Task 11 — Add dimensions

To fully define the sketch, we must add dimensions to position the slots and define their size. Even though we only drew one of the slots, the dimensions can be on any of the slots.

1 Add dimensions to the upper right slot as shown.

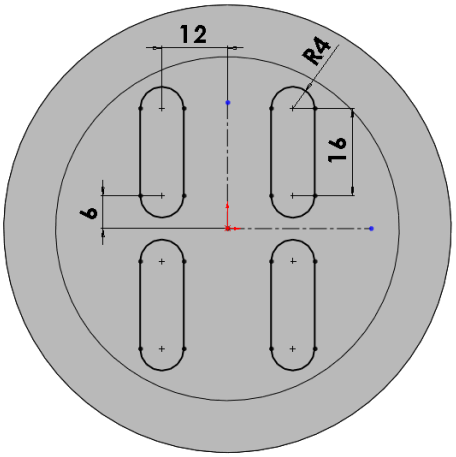
To add the **16mm** dimension, select the two arcs, not the vertical line.



2 Add dimensions as shown to the upper left slot to position it.

Both of these dimensions go from the lower arc to one of the centerlines.

3 Fully defined. The sketch geometry should now be all black, showing that the sketch is fully defined.



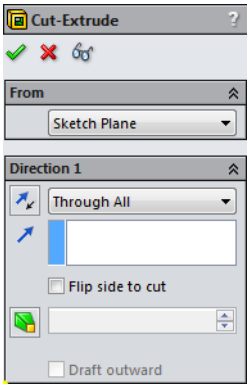
Task 12 — Create the cuts

The four slots must cut completely through the `Binding Anchor`. When we create the cut, it must be done so that if we need to change the thickness of the material later in the design process, the slots do not have to be redone.

1 Click **Insert, Cut, Extrude** from the menu.

2 Click  on the **View** toolbar to change the view to `Isometric`.

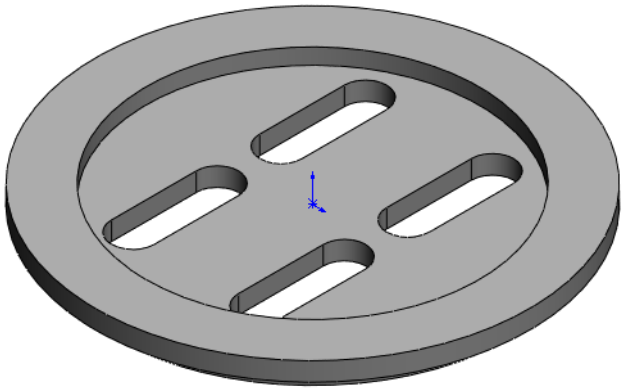
3 From the list in the PropertyManager select **Through All**.



Through All

The end condition **Through All** will make the cut go through all the geometry. If, when we later analyze the part for strength we determine that it needs to thicker, we will not have to redo the slots because they will go through the entire part, no matter how thick it is.

4 Complete the cut. Click ✔.



Renaming Features

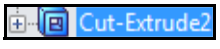
All the features shown in the FeatureManager design tree can be renamed. Renaming features can make them easier to locate as the parts become more complex.

There are three methods to rename features:

- **Click-Pause-Click. Click** on the feature name, **Pause**, **Click** the name again, type the new name.
- Click on the feature name, press **F2**, type the new name.
- Right-click the feature name and select **Properties**. Change the name in the Properties dialog box.

Task 13 — Rename the slots

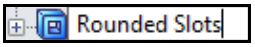
1 In the FeatureManager design tree, click once on `Cut-Extrude2`, this is the slots we just created.



2 Press **F2**. The feature name now has a box around it and a flashing cursor.



3 Type `Rounded Slots` for the new name.



4 The feature has now been renamed to something more descriptive.

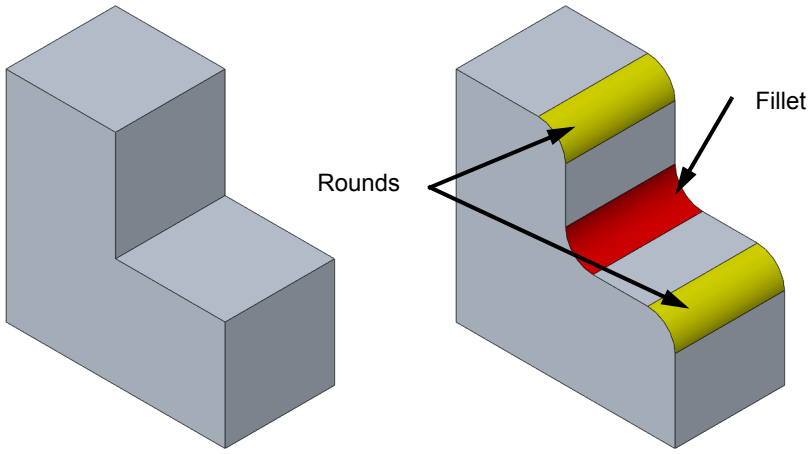


Notice that the feature's icon does not change. The  shows that this feature is a **Cut-Extrude**.

Filleting

Filleting refers to both fillets and rounds. The distinction is made by the geometric conditions, not the command itself. Fillets are created on selected edges. Those edges can be selected in several ways.

Both fillets (adding volume), shown in red, and rounds (removing volume), shown in yellow, are created with this command. The orientation of the edge or face determines which is used.

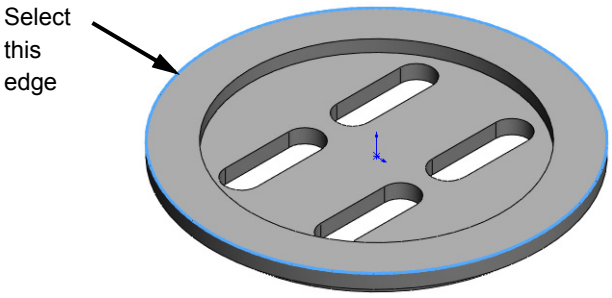


Task 14 — Round an outside corner

All the existing edges in this model are sharp. To meet our design intent, all the exposed edges need to be rounded.

1 Click **Fillet** on the **Features** toolbar.

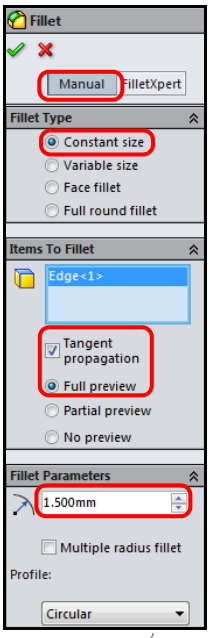
2 Select the edge shown.



3 Type **1.5mm** for the fillet radius.

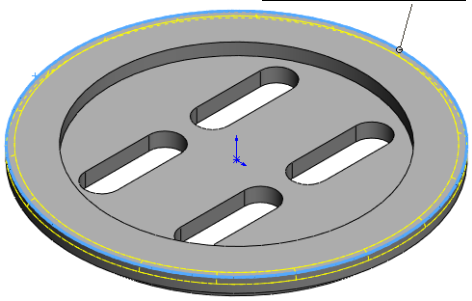
4 The following should be set by default. If not, change them as follows:

- **Manual**
- Fillet Type - **Constant radius**
- Tangent propagation - **Selected**
- Full preview - **Selected**



5 Preview. Once **Full preview** is selected, the outline of the fillet will be shown in yellow.

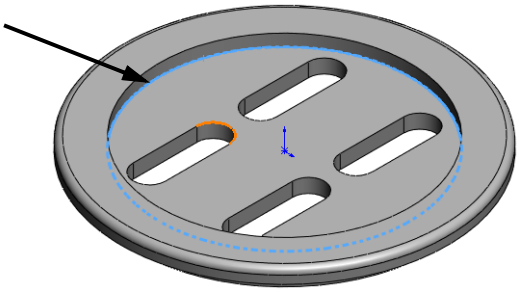
The fillet radius will be shown in a callout, Radius: 1.5mm attached to the edge. Click ✓.



Task 15 — Add a fillet to the inside edge

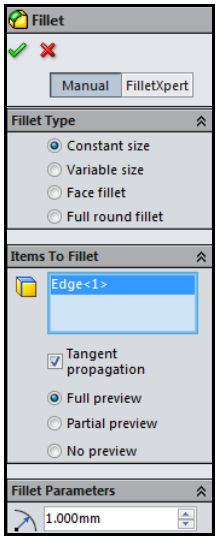
1 Click **Fillet** on the **Features** toolbar.

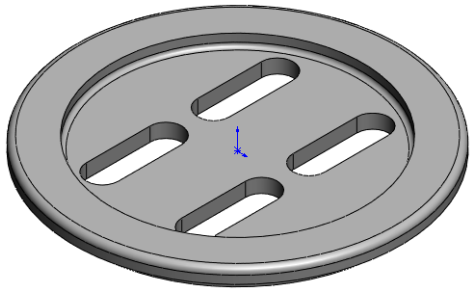
2 Select the edge shown.



3 Type **1mm** for the fillet radius.

4 Click ✓. The inside edge now has a fillet.





Editing Features

After a feature is created, it can easily be changed. For the extrudes and cuts we made, each end condition or depth could be changed to reflect changes in our design intent, or changes required by later analysis.

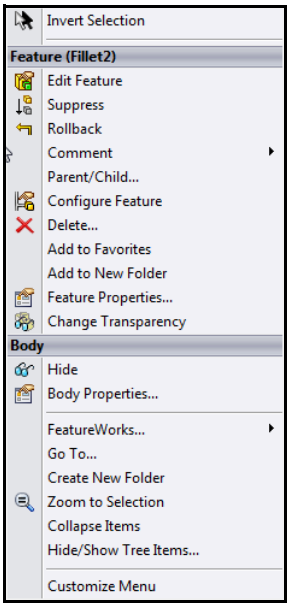
For the fillet just created, we could add additional edges to the feature or change the fillet radius.

To edit a feature, right-click the feature either in the graphics area or FeatureManager design tree and select **Edit Feature**.

Task 16 — Edit the Fillet

1 In the FeatureManager design tree, right-click the feature `Fillet2` and select **Edit Feature** from the Context toolbar.

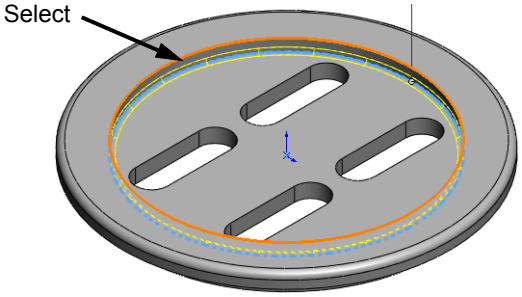




SolidWorks can be customized to show the Context toolbar with a text menu below it or just a text menu.

The PropertyManager will now show the same information as when we first applied the fillet.

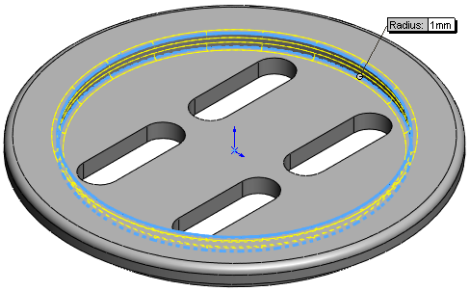
2 Select the edge shown.



Note: One of these edges will require a round while the other will require a fillet. Both are being done in the same command.

3 Once selected, the preview will show that the edge will be rounded as part of the `Fillet2` feature. Click ✔.

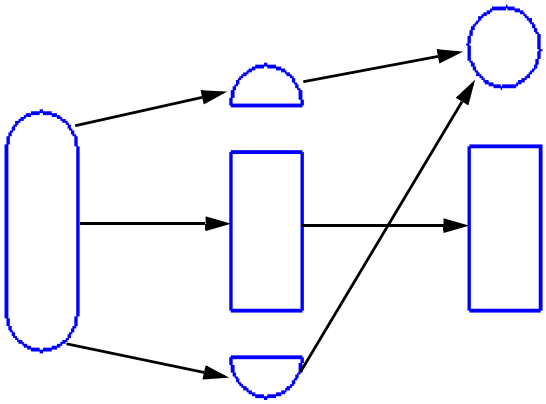
4 **Save** the part.



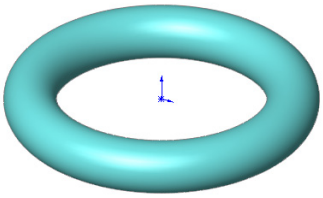
How much does it weight now?

The same principle used earlier of adding and subtracting volumes still applies, however it is now more complicated.

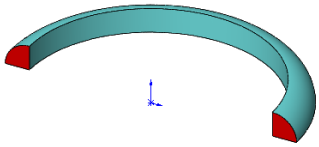
The volume removed by the slots is not too difficult as the area of each slot can be thought of as a rectangle and circle

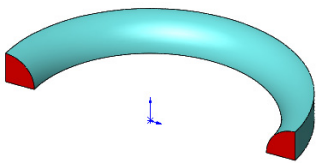


The fillets are more complicated. The two rounded corners are each part of a torus (donut).

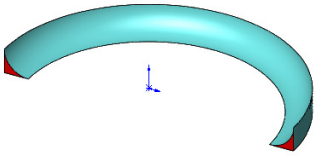


The volumes of the rounds are parts of the torus. The section views at the right show what the two would look like. The volumes are ***not*** one-quarter of the volume of the torus. The equations to determine their volumes are available in both engineering and mathematics handbooks.





The inside corner fillet is even more complicated but still solvable by looking up the equation.



Using SolidWorks To Get The Weight

Rather than manually solve for the volume of our part and lookup the density of the material, SolidWorks provides tools to solve for the volume and weight of the part.

Add Material

SolidWorks provides a library of materials that can be assigned to parts. Once a material is applied, it will be used by SolidWorks to calculate the weight of the part.

Adding material to a part also changes its visual properties (what it looks like) and graphic properties like the crosshatch used in drawings.

The material assigned to the part can also be used for stress analysis and for photorealistic rendering.

To assign a Material to a part:

- Click  on the **Standard** toolbar.
- Click **Edit, Appearance, Material** in the menu.
- Right-click **Material** in the FeatureManager design tree and select **Edit Material**.

Task 17 — Add material to the part

We are going to manufacture this part from Aluminum 2015.

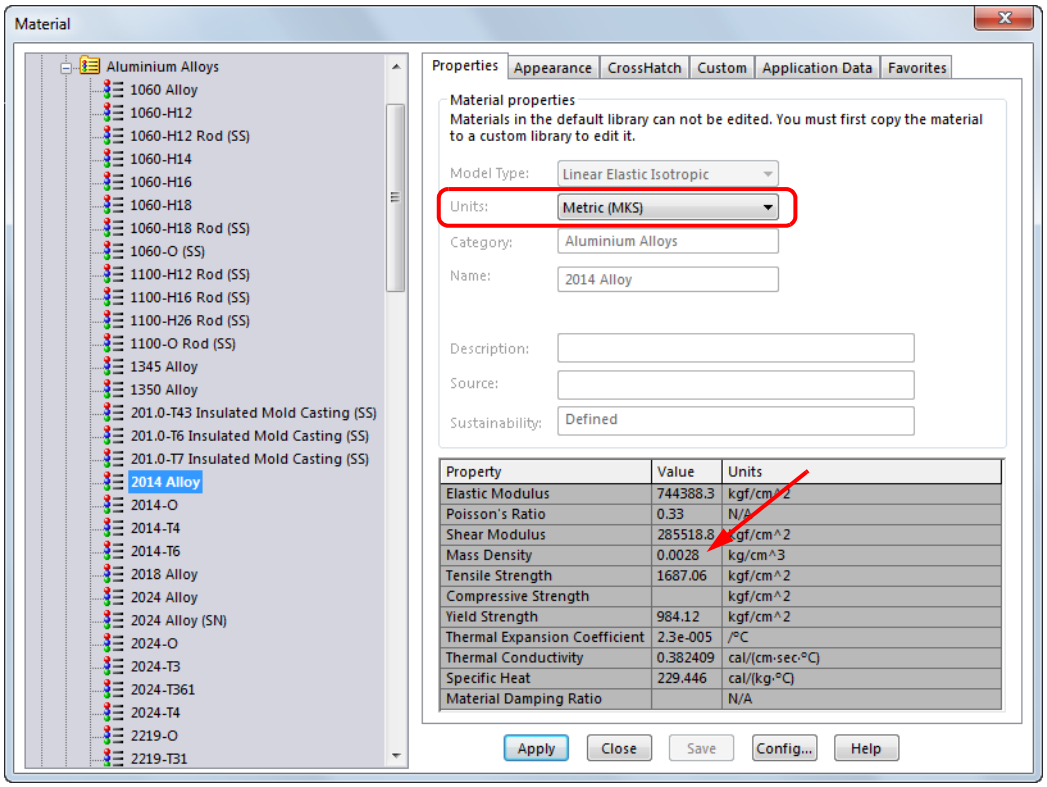
1. Click **Edit, Appearance, Material** from the menu.

   The **Materials Editor** will open.
2. Click the Plus sign next to **Aluminum Alloys** to expand the list.
3. Select **2015 Alloy**.

   Select **Metric (MKS)** for **Units**.

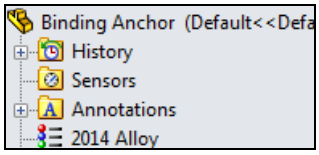
Examine the Physical Properties. Density is listed as .0028 g/mm^3. This is the same value we determined earlier.

Note: Certain graphics cards support RealView advanced graphics visual properties. To find out if yours does, consult the **Help** documentation inside SolidWorks.



4 Click **Apply** and then **Close** to apply the material.

The material is now listed in the FeatureManager design tree.



Mass Properties

Physical properties of a part can easily be calculated using the **Mass Properties** tool. This tool will not only calculate the volume and weight of the part, but many other properties needed during the design and analysis of a part.

To calculate Mass properties:

- Click **Tools, Mass Properties** from the menu
- Or, click  on the Tools toolbar

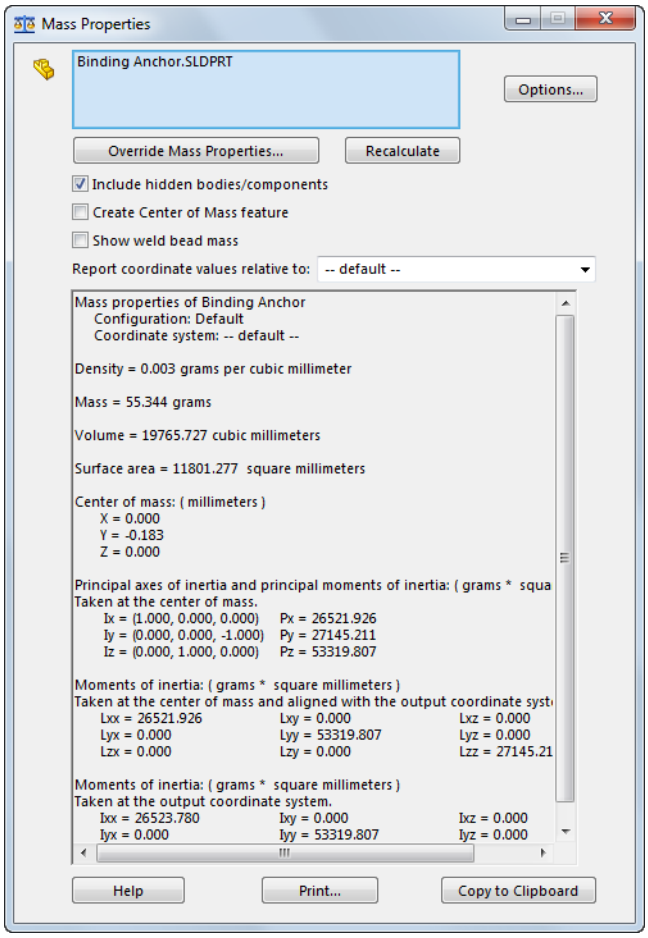
Task 18 — Determine The Weight

1 Click **Tools, Mass Properties** from the menu.

The Mass Properties box will appear.

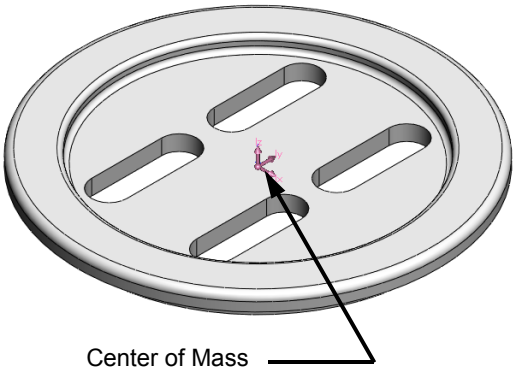
The Volume is calculated to be **19,765.727mm^3** and the Mass is **55.344 grams**.

When we calculated the weight earlier it was 62.68 grams. This was before we removed material with the four slots and two rounds.



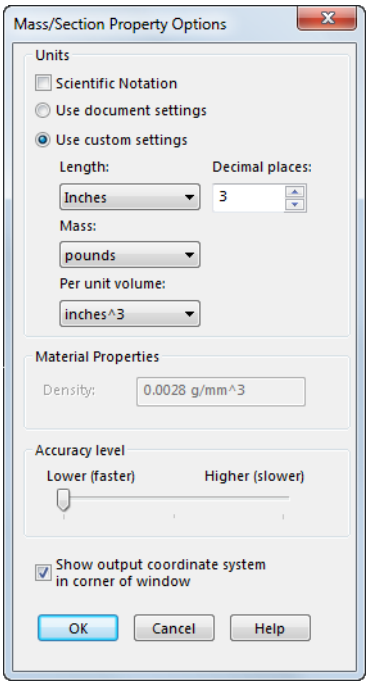
2 The Center of Mass is the balance point of the part. If we could suspend the part at this point, it would not want to tip over.

The center of mass is displayed numerically in the box and graphically by a pink triad.



Change Units.

The units for the `Binding Anchor` part are in millimeters, grams, seconds, so the mass properties displayed in millimeters and grams. If we need the mass properties in different units, such as inch, pound, second, the conversion is simple.

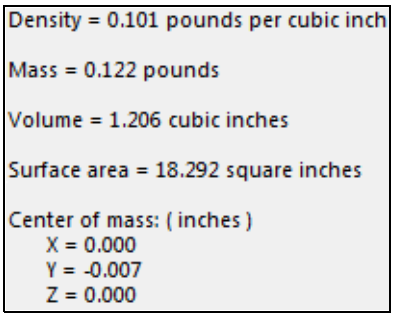


3 Click the **Options** button.

4 Select **Use custom settings**. Select:

- Length - **Inches**
- Mass - **Pounds**
- Per unit volume - **inches^3**

5 Click **OK**.



Note: The units have only been changed in this output. The part still uses millimeters as the unit of length

6 Click **Close** to close the Mass Properties.

7 **Save** the part.